

Linear Regression

Regression analysis is a statistical technique for determining the relationship between two or more than two variables. There are two types of variables in regression analysis

1. Independent variable
2. Dependent variable.

Independent variables are also known as predictor variables. These are the variables that do not change. On the other side, the variables whose values change are known as dependent variables. These variables depend on the independent variables. Dependent variables are also known as response variables.

Linear regression carry out a statistical procedure to predict the response variable based on the input of the independent or predictor variables.

Linear Regression is of the following two types:

1. Simple Linear Regression – Based on the value of the single explanatory variable, the value of the response variable changes .
2. Multiple Linear Regression – The value is dependent upon more than one explanatory variables in case of multiple linear regression.

Simple Linear Regression

Simple linear regression is used for finding the relationship between the dependent variable Y and the independent or predictor variable X. Both of these variables are continuous in nature

The equation of a simple linear regression model to calculate the value of the dependent variable, Y based on the predictor X is as follows:

$$y_i = \beta_o + \beta_1 x + \epsilon$$

Y_i is calculated with the input variable X_i for every i^{th} data point;

The coefficients of regressions are denoted by β_0 and β_1 ;

The i^{th} value of x has ϵ_i as its error in the measurement.

Multiple Linear Regression in R

In many cases, there may be possibilities of dealing with more than one predictor variable for finding out the value of the response variable. Therefore, the simple linear models cannot be utilised as there is a need for undertaking multiple linear regression for analyzing the predictor variables.

Using the two explanatory variables, we can delineate the equation of multiple linear regression as follows:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

The two explanatory variables X_{1i} and X_{2i} , determine Y_i , for the i^{th} data point

the predictor variables are also determined by the three parameters β_0 , β_1 , and β_2 of the model, and by the residual ϵ_i of the point i from the fitted surface.

Example

Creating a Linear Regression in R

When you calculate the age of a child based on their height, you are assuming the older they are, the taller they will be.

In this case, linear regression assumes that there exists a linear relationship between the response variable and the explanatory variables. This means that you can fit a line between the two (or more variables). In this example it is clear that there is a relationship between the age of children and their height.

$$\text{Height} = a + \text{Age} * b$$

A linear regression can be calculated in R with the command `lm`

`lm` function in R:

The `lm` command takes the variables in the format:

`lm([target] ~ [predictor / features], data = [data source])`

```
Call:
lm(formula = height ~ age, data = ageandheight)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27238 -0.24248 -0.02762  0.16014  0.47238

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  64.9283    0.5084  127.71  < 2e-16 ***
age          0.6350    0.0214   29.66 4.43e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.256 on 10 degrees of freedom
Multiple R-squared:  0.9888,    Adjusted R-squared:  0.9876
F-statistic: 880 on 1 and 10 DF,  p-value: 4.428e-11
```

For Multiple Linear Regression

$$\text{Height} = a + \text{Age} \times b_1 + (\text{Number of Siblings}) \times b_2$$

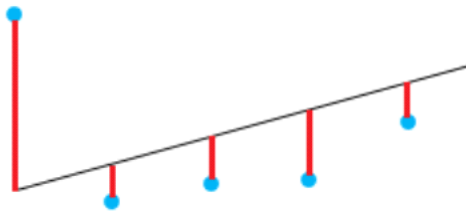
You are now looking at the height as a function of the age in months and the number of siblings the child has

1. Coefficients

In the red square, you can see the values of the intercept (“a” value) and the slope (“b” value) for the age. These “a” and “b” values plot a line between all the points of the data. So, in this case, if there is a child that is 20.5 months old, a is 64.92, and b is 0.635, the model predicts (on average) that its height in centimeters is around $64.92 + (0.635 * 20.5) = 77.93$ cm.

2. Residuals

A good way to test the quality of the fit of the model is to look at the residuals or the differences between the real values and the predicted values. The straight line in the image above represents the predicted values. The red vertical line from the straight line to the observed data value is the residual.



The idea here is that the sum of the residuals is approximately zero or as low as possible. In real life, most cases will not follow a perfectly straight line, so residuals are expected

3. How to Test if your Linear Model has a Good Fit

One measure very used to test how good your model is is the coefficient of determination or R^2 . This measure is defined by the proportion of the total variability explained by the regression mode

$$R^2 = \frac{\text{Explained Variation of the model}}{\text{Total variation of the mode}}$$

For models that fit the data well, R^2 is near 1. Models that poorly fit the data have R^2 near 0. In the examples below, the first one has an R^2 of 0.02; this means that the model explains only 2% of the data variability. The second one has an R^2 of 0.99, and the model can explain 99% of the total variability

4. P Value

```
Call:
lm(formula = height ~ age + no_siblings, data = ageandheight)

Residuals:
    Min       1Q   Median       3Q      Max
-0.28029 -0.22490 -0.02219  0.14418  0.48350

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  64.95872    0.55752  116.515 1.28e-15 ***
age          0.63516    0.02254   28.180 4.34e-10 ***
no_siblings  -0.01137    0.05893   -0.193  0.851
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2693 on 9 degrees of freedom
Multiple R-squared:  0.9888,    Adjusted R-squared:  0.9863
F-statistic: 397.7 on 2 and 9 DF,  p-value: 1.658e-09
```

A standard way to test if the predictors are not meaningful is by looking if the p-values smaller than 0.05.

The p-value for age is 4.34×10^{-10} or 0.000000000434. A very small value means that age is probably an excellent addition to your model.

The p-value for the number of siblings is 0.85. In other words, there's an 85% chance that this predictor is not meaningful for the regression.